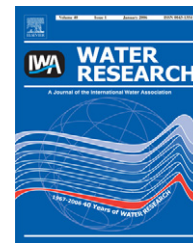


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Performance of single chamber biocatalyzed electrolysis with different types of ion exchange membranes

René A. Rozendal^{a,b}, Hubertus V.M. Hamelers^{a,*}, Redmar J. Molenkamp^b, Cees J.N. Buisman^{a,b}

^aSub-Department of Environmental Technology, Wageningen University, Bomenweg 2, P.O. Box 8129, 6700 EV Wageningen, The Netherlands

^bWetsus, centre for sustainable water technology, Agora 1, P.O. Box 1113, 8900 CC Leeuwarden, The Netherlands

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ABSTRACT

In this paper hydrogen production through biocatalyzed electrolysis was studied for the first time in a single chamber configuration. Single chamber biocatalyzed electrolysis was tested in two configurations: (i) with a cation exchange membrane (CEM) and (ii) with an anion exchange membrane (AEM). Both configurations performed comparably and produced over $0.3 \text{ m}^3 \text{ H}_2/\text{m}^3$ reactor liquid volume/day at 1.0 V applied voltage (overall hydrogen efficiencies around 23%). Analysis of the water that permeated through the membrane revealed that a large part of potential losses in the system were associated with a pH gradient across the membrane (CEM $\Delta \text{pH} = 6.4$; AEM $\Delta \text{pH} = 4.4$). These pH gradient associated potential losses were lower in the AEM configuration (CEM 0.38 V; AEM 0.26 V) as a result of its alternative ion transport properties. This benefit of the AEM, however, was counteracted by the higher cathode overpotentials occurring in the AEM configuration (CEM 0.12 V at 2.39 A/m^2 ; AEM 0.27 V at 2.15 A/m^2) as a result of a less effective electroless plating method for the AEM membrane electrode assembly (MEA).

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1. Introduction

1.1. Biocatalyzed electrolysis

Recently, it has been demonstrated that biocatalyzed electrolysis (or the bio-electrochemically assisted microbial reactor (BEAMR) process) is an interesting new technology for the production of hydrogen gas from wastewaters (Liu et al., 2005b; Rozendal and Buisman, 2005; Rozendal et al., 2006b). Biocatalyzed electrolysis is related to microbial fuel cell (MFC) technology as it also applies electrochemically active micro-organisms (Logan et al., 2006). Electrochemically active micro-organisms are capable of using an electrode surface as an external electron acceptor for the oxidation of organic compounds. As a result this electrode functions as a

biological anode. At pH 7, electrochemically active micro-organisms on the biological anode typically oxidize organic compounds at a potential of about -0.20 to -0.28 V (Logan et al., 2006). In biocatalyzed electrolysis, this biological anode is coupled to a hydrogen evolution reaction (HER; -0.42 V at pH 7) at the cathode, which results in an electromotive force (emf) of about -0.14 to -0.22 V (= cathode potential–anode potential) for the biocatalyzed electrolysis process. The negative value of the emf implies that a voltage of 0.14 to 0.22 V needs to be applied. Electrical energy, therefore, needs to be invested in order to generate hydrogen gas at the cathode. Compared to conventional water electrolysis, however, this energy investment is only low, as conventional water electrolysis theoretically requires applied voltages above 1.23 V (Kinoshita, 1992).

*Corresponding author. Tel.: +31 0 317 483447; fax: +31 0 317 482108.

E-mail addresses: rene.rozendal@wetsus.nl (R.A. Rozendal), bert.hamelers@wur.nl (H.V.M. Hamelers), r.j.molenkamp@student.rug.nl (R.J. Molenkamp), cees.buisman@wur.nl (C.J.N. Buisman).

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1.2. Performance

Previously we studied the performance of a two-chamber biocatalyzed electrolysis setup at an applied voltage of 0.5 V (Rozendal et al., 2006b). At this applied voltage the volumetric hydrogen production rate averaged $0.02 \text{ m}^3 \text{ H}_2/\text{m}^3$ reactor liquid volume/day. Detailed analysis of the results revealed that the performance was to a large extent limited by inefficiencies occurring in the cathode reaction. In theory, however, HER cathodes should be able to outperform biological anodes up to 4 orders of magnitude. For example, at an overpotential of 0.025 V biological anodes can achieve current densities in the order of 1 A/m^2 , while HER cathodes at the same overpotential are able to exceed $10,000 \text{ A/m}^2$ (Kinoshita, 1992). In our previous study, however, the cathode overpotential was found to be already 0.28 V at a current density of 0.5 A/m^2 , while the biological anode overpotential was 0.04 V. This clearly leaves room for improvement of the cathode performance. We estimate that if these improvements of the cathode performance can be realized, biocatalyzed electrolysis systems can be optimized to produce over $10 \text{ m}^3 \text{ H}_2/\text{m}^3$ reactor liquid volume/day (Rozendal et al., 2006b). These rates, however, also require an optimized geometrical design of the biocatalyzed electrolysis system.

1.3. Single chamber configuration

A better geometrical design of biocatalyzed electrolysis system can be achieved by changing the system from a configuration with two liquid chambers (i.e. two-chamber configuration) to a configuration with only one liquid chamber (i.e. single chamber configuration) by implementing a gas diffusion electrode. When this modification can be achieved without affecting the electrochemical performance, the volumetric hydrogen production rate theoretically doubles. Single chamber configurations have been tested successfully for the MFC (Liu et al., 2004; Sell et al., 1989). Interestingly, Liu and Logan demonstrated that the performance of single chamber MFCs can be significantly increased (494 versus 262 mW/m^2) by removing the cation exchange membrane (CEM) from the system (Liu and Logan, 2004).

In biocatalyzed electrolysis, however, the presence of a membrane is essential for the purity of the hydrogen that is produced at the cathode. Without the membrane the produced hydrogen will be polluted with gaseous metabolic products from the anode chamber (e.g. CO_2 , CH_4 , H_2S). Furthermore, a significant amount of the produced hydrogen will be lost through diffusion to the anode chamber. A single chamber biocatalyzed electrolysis configuration, therefore, can only be successful if it functions optimally with a membrane included in the system.

1.4. Membrane ion transport

Previously it was shown in two-chamber MFCs that the performance loss associated with the CEM is caused by the transport of cation species other than protons (Na^+ , K^+ , NH_4^+ , Ca^{2+} , and Mg^{2+}) through the membrane (Rozendal et al., 2006a; Zhao et al., 2006). At a typical wastewater pH of 7 the proton concentration is about 10^{-4} mM . The concentrations of

cation species other than protons in wastewater, however, are typically 4–5 orders of magnitude higher. As a result electroneutrality in the MFC is mainly observed by the transport of cation species other than protons through the CEM. Protons, however, are consumed equimolarly with electrons at the cathode in both the oxygen reduction reaction in MFCs ($\text{O}_2 + 4 \text{ H}^+ + 4 \text{ e}^- \rightarrow 2 \text{ H}_2\text{O}$) as in the HER in biocatalyzed electrolysis ($2 \text{ H}^+ + 2 \text{ e}^- \rightarrow \text{H}_2$) and the transport of cation species other than protons, therefore, causes a pH increase in the cathode chamber (Gil et al., 2003; Rozendal et al., 2006a; Zhao et al., 2006). This pH increase negatively affects the system performance, as the Nernst equation (Bard and Faulkner, 2001) states that for every pH unit difference between anode and cathode created in this way, an extra potential loss of about 0.06 V will occur in the system. Similar effects also occur in single chamber configurations.

1.5. Anion exchange membrane

A possible solution to the pH gradient associated potential losses is the application of an anion exchange membrane (AEM) instead of a CEM (Fig. 1). In the case of an AEM, electroneutrality is observed not by the transport of cations from the anode to the cathode, but by the transport of anions from the cathode to the anode. For biocatalyzed electrolysis this implies that hydrogen at the cathode is not produced from the reduction of protons, but from the reduction of water that diffuses through the membrane from anode to cathode. In this reaction hydroxyl ions are produced equimolarly with the consumed electrons ($2 \text{ H}_2\text{O} + 2 \text{ e}^- \rightarrow 2 \text{ OH}^- + \text{H}_2$), which are then transported from cathode to anode through the AEM in order to observe electroneutrality. We predict that this has a neutralizing effect on the pH gradient across the membrane.

1.6. Objective

The objective of this study is twofold: (i) to study the performance of single chamber biocatalyzed electrolysis; and (ii) to study the influence of the type of ion exchange membrane on the performance. For this purpose we have operated two single chamber biocatalyzed electrolysis configurations in parallel, one with a CEM and one with an AEM.

2. Materials and methods

2.1. Electrochemical cell

The experiments were performed in two identical electrochemical cells as previously described (Rozendal et al., 2006b). Although the electrochemical cells had two chambers, they were operated in a single chamber configuration, in which only the anode chamber (liquid volume 3.3 L) was filled with medium (see below) and the cathode chamber (volume 3.3 L) functioned as a gas collection chamber. In one electrochemical cell the chambers were separated by a CEM (Nafion[®] 117, surface area 256 cm^2); in the other by an AEM (Fumasep[®] FAB, FuMA-Tech GmbH, surface area 256 cm^2). The anode consisted of graphite felt (surface area 400 cm^2 , thickness

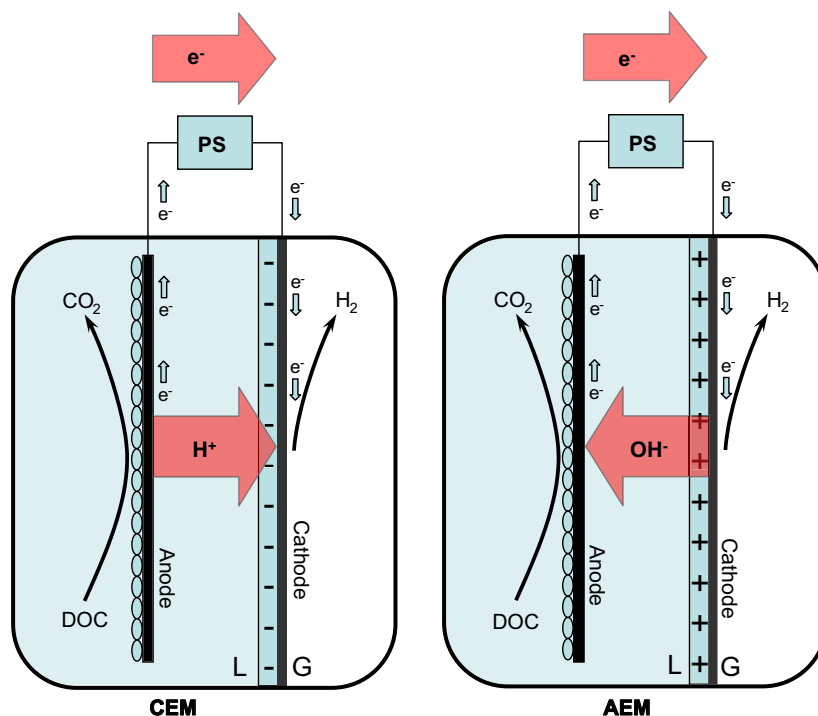


Fig. 1 – Schematic representation of single chamber biocatalyzed electrolysis with a cation exchange membrane (CEM) and an anion exchange membrane (AEM). DOC = dissolved organic compounds, PS = power supply, L = liquid phase, G = gas phase.

3 mm—FMI Composites Ltd., Galashiels, Scotland); the cathode consisted of platinum and was integrated with the membrane as a membrane electrode assembly (MEA; see below). The MEA was supported by a platinum coated (50 g/m^2) titanium mesh (surface area 400 cm^2 , thickness 1 mm, specific surface area $1.7\text{ m}^2/\text{m}^2$ —Magnetos special anodes BV, Schiedam, The Netherlands), which functioned both as part of the cathode and as the current collector. The anode and cathode were electrically connected to an adjustable power supply (ES 030-5, Delta Elektronika BV, The Netherlands). The current through the electrical circuit was determined (Ohm's law) by incorporating a high-precision shunt resistor ($10\ \Omega$) into the electrical circuit. The adjustable power supply was equipped with a sensing lead to automatically correct for the voltage loss across the shunt resistor. The anode chamber was equipped with an Ag/AgCl reference electrode (+0.2 V against NHE). A pH electrode was placed externally to the cell in a flow cell through which the anode medium was continuously pumped at a rate of 250 mL/min. The anode chamber was operated at pH 7 (Liquisys M CPM 253, Endress+Hauser). The electrochemical cells were connected to a data logger (Ecograph T, Endress+Hauser), which continuously logged the applied voltage, current, anode potential, cathode potential, pH, temperature and gas production.

2.2. MEA preparation

The MEAs were prepared by means of electroless plating. The electroless plating method used for preparing the CEM MEA

was based on a method developed by Millet et al. (1990). One side of the membrane was subjected to two cycles of exposure for 15 min to a 200 mL of an aqueous $Pt(NH_3)_4Cl_2$ -solution (2 mg Pt/mL) and subsequent exposure for 2 h to 500 mL of an aqueous solution of sodium borohydride ($NaBH_4$; 15 g/L). This method is based on the diffusion of the positively charged platinum complex $[Pt(NH_3)_4]^{2+}$ into the CEM and subsequently reducing this complex in situ to metallic platinum with sodium borohydride. This plating method results in a MEA with a conducting platinum layer outside the membrane in electrical contact with catalytic platinum microparticles inside the membrane. According to Millet et al. (1990) this procedure results in a MEA with a platinum loading of about 1.0 mg Pt/cm^2 .

For the AEM the plating method was adapted by replacing the aqueous $Pt(NH_3)_4Cl_2$ -solution with an aqueous K_2PtCl_6 -solution (2 mg Pt/mL). This solution contains the negatively charged complex $[PtCl_6]^{2-}$ that can diffuse into the AEM.

2.3. Experimental procedures and calculations

All experiments were performed at 303 K. The anode chambers of the electrochemical cells were filled and continuously fed (4 mL/min) with anode medium containing (in deionized water): 1.36 g/L $NaCH_3COO \cdot 3H_2O$, 0.74 g/L KCl, 0.58 g/L NaCl, 0.68 g/L KH_2PO_4 , 0.87 g/L K_2HPO_4 , 0.28 g/L NH_4Cl , 0.1 g/L $MgSO_4 \cdot 7H_2O$, 0.1 g/L $CaCl_2 \cdot 2H_2O$ and 1 mL/L of a trace element mixture (Zehnder et al., 1980). The electrochemical cells were started up by inoculation with 200 mL effluent taken from an active bio-electrochemical cell (Rozendal et al.,

2006a). In total the cells have been operated for about 100 days. After stabilization at an applied voltage of 1.0 V the electrochemical cells were evaluated on the basis of three techniques: (i) applied voltage scans; (ii) electrochemical impedance spectroscopy (EIS); and (iii) hydrogen yield tests. During the applied voltage scans and EIS, the gas collection chamber was flushed continuously with water saturated nitrogen gas (99.999% N₂, air products) in order to keep the experimental conditions constant. During the hydrogen yield tests the gas collection chamber was not flushed with nitrogen gas.

The applied voltage scans were obtained in triplicate by means of chronoamperometry. For this purpose the electrochemical cells were connected to a potentiostat (Autolab PGSTAT12, Eco Chemie B.V., The Netherlands). The applied voltage was increased stepwise from 0 to 1 V. The current generation was logged every 10 s for 1 h at 0, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8, and 1.0 V. The last 10 data points of every applied voltage step were averaged and plotted in the applied voltage scan.

For the EIS the electrochemical cells were connected to a potentiostat equipped with a frequency response analyzer (Autolab PGSTAT30, Eco Chemie B.V., The Netherlands). EIS was performed at an applied voltage of 1.0 V with an amplitude of 10 mV in the frequency range of 5 mHz to 100 kHz.

The hydrogen yield tests lasted for 48 h and were repeated in duplicate. In order to calculate the cumulative acetate consumption the anode was regularly sampled and analyzed for acetate content using ion chromatography (Metrohm 761 Compact IC equipped with a conductivity detector and a Metrosep Organic Acids 6.1005.200 ion exclusion column). At the start of the hydrogen yield tests the gas collection chamber contained only nitrogen gas. During the hydrogen yield tests the gas collection chamber was connected to a gas flow meter (Milligascounter[®], Ritter) for quantifying the total gas production. The gas collection chamber was sampled regularly and analyzed for its hydrogen fraction with a gas chromatograph (Shimadzu GC-2010 equipped with a thermal conductivity detector and a Varian molsieve 5A column). The actual hydrogen gas production was calculated from the total gas production and the measured hydrogen fractions by means of a mass balance equation adapted from (Logan et al., 2002):

$$V_{H,t} = V_{H,t-1} + (V_{T,t} - V_{T,t-1}) \frac{(C_{H,t} + C_{H,t-1})}{2} + V_{cat}(C_{H,t} - C_{H,t-1}) \quad (1)$$

with $V_{H,t}$ and $V_{H,t-1}$ the cumulative hydrogen gas production in liters (L) on sample time t and previous sample time $t-1$, respectively, $V_{T,t}$ and $V_{T,t-1}$ the total measured gas production in liters (L) on sample time t and previous sample time $t-1$, respectively, $C_{H,t}$ and $C_{H,t-1}$ the measured hydrogen fractions on sample time t and previous sample time $t-1$, respectively, and V_{cat} the gas collection chamber volume (3.3 L). The expected hydrogen production based on the cumulative charge production was calculated according to

$$V_{HC,t} = \frac{Q_t V_M}{2F} \quad (2)$$

with $V_{HC,t}$ the expected hydrogen production based on the cumulative charge production in liters (L) on sample time t , Q_t the cumulative charge production (equal to the integrated current over time) in coulombs (C) on sample time t , F Faraday's number (96485.3 C/mol), and V_M the molar gas volume (25.2 L/mol at 303 K).

During the hydrogen yield tests small amounts of water permeated through the membrane from the anode to the cathode side of the membrane into the gas collection chamber. This permeated water was collected from the gas collection chamber at the end of the hydrogen yield tests and analyzed for pH, conductivity and ion content. Carbonate concentrations were determined using a total organic carbon analyzer (Shimadzu TOC-VCPH). Other anion concentrations (Cl^- , NO_2^- , NO_3^- , PO_4^{3-} , and SO_4^{2-}) were determined using an ion chromatograph (Metrohm 761 Compact IC) equipped with a conductivity detector and an anion column (Metrosep A Supp 5 6.1006.520). Ammonium concentrations were photometrically determined using standardized test kits (ammonium cuvette test LCK303, XION 500 spectrophotometer, Dr. Lange Nederland B.V., The Netherlands). Other cation concentrations (Na^+ , K^+ , Ca^{2+} , and Mg^{2+}) were determined using inductively coupled plasma-optical emission spectroscopy (ICP-OES—Perkin Elmer Optima 3000XL). The ion content data of the permeated water were used to calculate transport numbers (t) for protons and hydroxyl ions through the ion exchange membranes. Because in practice no distinction can be made between the membrane transport of protons from anode to cathode and the membrane transport of hydroxyl ions from cathode to anode, the following transport number was defined:

$$t_{H/OH} = \frac{Q_{H/OH}}{Q_{total}} = \frac{Q_{total} - Q_{other}}{Q_{total}} \quad (3)$$

with $t_{H/OH}$ the transport number for the membrane transport of protons from anode to cathode and hydroxyl ions from cathode to anode, $Q_{H/OH}$ the membrane transport of positive charge from anode to cathode (or negative charge from cathode to anode) in the form of protons and hydroxyl ions, expressed in coulombs (C), Q_{other} the membrane transport of positive charge from anode to cathode (or negative charge from cathode to anode) in the form of ions other than proton and hydroxyl ions, expressed in coulombs (C), and Q_{total} the total membrane transport of positive charge from anode to cathode (or negative charge from cathode to anode), expressed in coulombs (C). As the electroneutrality principle states that the total membrane transport of positive charge from anode to cathode (or negative charge from cathode to anode) is equal to the total electron transport from anode to cathode through the electrical circuit, Q_{total} equals the integrated current over time.

After the experiments scanning electron microscope photos were taken of a cross-section of the MEAs. Prior to taking these scanning electron microscope photos, the plated membranes were submerged in liquid nitrogen to make them brittle. Subsequently, the membranes were broken and SEM photos were taken on the line of breakage. For the SEM photos a JEOL JSM-6480LV SEM was used (acceleration voltage 20 kV, LV-mode 60 Pa, BEI detector). The microscope was equipped

with a NORAN System SIX Model 300 X-ray microanalysis system (Thermo Electron Corporation) for element analysis (linescan mode).

3. Results and discussion

3.1. Applied voltage scans

Fig. 2a gives the applied voltage scans of the electrochemical cells taken after stabilization of the performance at 1.0V. A comparison between the electrochemical cells shows that especially in the lower applied voltage range the CEM configuration outperforms the AEM configuration. The CEM configuration already starts to produce current at applied voltages above 0.2V, while the AEM configuration only starts to produce current at applied voltages above 0.4V. The theoretical applied voltage required for production of hydrogen from acetate at the experimental conditions is around 0.17V.

When the voltage comes above 0.6V for both the CEM and the AEM configuration, the lines rapidly become steeper. The average slope of the applied voltage scan of the CEM configuration increases from $1.4\text{ A/m}^2/\text{V}$ between 0.5 and 0.6V to $5.9\text{ A/m}^2/\text{V}$ between 0.8 and 1.0V, while the average slope of the applied voltage scan of the AEM configuration even increases from $1.3\text{ A/m}^2/\text{V}$ between 0.5 and 0.6V to $6.6\text{ A/m}^2/\text{V}$ between 0.8 and 1.0V. As a result of this, the AEM configuration at 1.0V even slightly outperforms CEM configuration, even though it was performing worse at the lower applied voltages.

Fig. 2b shows the electrode potentials measured during the applied voltage scans. Because in a single chamber configuration only the anode chamber can be equipped with a reference electrode, only the anode potential can be measured as a true electrode potential. The cathode potential can only be measured through the membrane and therefore also includes the membrane potential.

Electrode overpotentials (i.e. potential losses) can be calculated by comparing the measured anode and cathode/

membrane potentials with theoretical potentials for acetate oxidation (-0.25 V) and proton reduction (-0.42 V). Fig. 2b shows that the performance of both single chamber biocatalyzed electrolysis configurations to a large extent was limited by the combined cathode/membrane overpotential. At an applied voltage of 1.0V (CEM $2.25 \pm 0.05\text{ A/m}^2$; AEM $2.37 \pm 0.04\text{ A/m}^2$) the anode overpotential was $0.12 \pm 0.001\text{ V}$ for the CEM configuration and $0.16 \pm 0.001\text{ V}$ for the AEM configuration, while the combined cathode/membrane overpotential was $0.72 \pm 0.001\text{ V}$ for the CEM configuration and $0.68 \pm 0.001\text{ V}$ for the AEM configuration. From Fig. 2b it can be seen that a large part of the cathode/membrane overpotentials already occurred below 0.5 A/m^2 . From the results it cannot unequivocally be determined which part of the cathode/membrane overpotential was caused by the cathode and which part was caused by the membrane. However, it seems unlikely that these losses could have been caused by the ohmic resistance of the membrane as these types of membranes typically have ohmic resistances in the order of $1\text{--}10\text{ }\Omega\text{cm}^2$ (Xu, 2005). Below an applied voltage of 0.6V the current densities in the cells were below 0.5 A/m^2 . This results in a maximum ohmic loss across the membrane in the order of 0.0005 V ($= 0.5 \times 10^{-4}\text{ A/cm}^2 \times 10\text{ }\Omega\text{cm}^2$), which is too low to explain the cathode/membrane overpotentials that were observed.

3.2. Hydrogen yield tests

Hydrogen yield tests were done to assess the hydrogen production potential of both configurations (Fig. 3). At first the hydrogen production was tested at an applied voltage of 0.5V, but the hydrogen production was too low to obtain reproducible results. Therefore, the applied voltage was increased to 1.0V. At this applied voltage the current density during the 48h hydrogen yield tests averaged $2.39 \pm 0.15\text{ A/m}^2$ for the CEM configuration and $2.15 \pm 0.26\text{ A/m}^2$ for the AEM configuration. These values are in the same range as predicted by the applied voltage scans (Fig. 2a), which shows that the applied voltage scans are indeed representative for

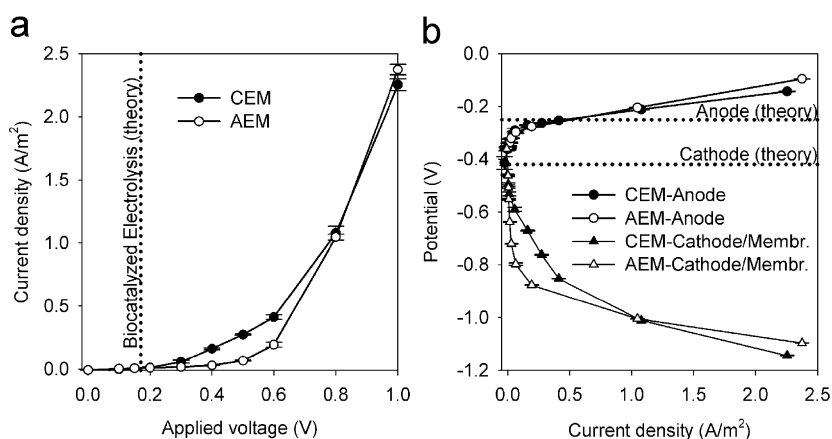


Fig. 2 – Evaluation of the performance of single chamber biocatalyzed electrolysis of acetate (10 mM) with a cation exchange membrane (CEM; Nafion[®] 117) and an anion exchange membrane (AEM; Fumasep[®] FAB): (a) applied voltage scans and (b) measured potentials during the applied voltage scans.

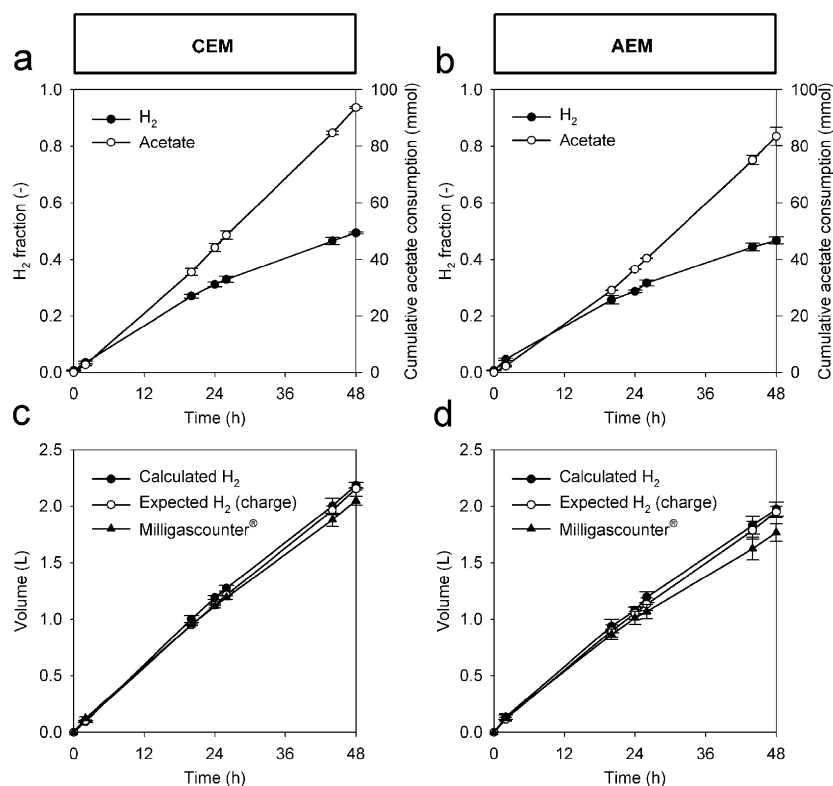


Fig. 3 – Experimental results of 48 h hydrogen yield tests of single chamber biocatalyzed electrolysis of acetate with a cation exchange membrane (CEM; Nafion[®] 117) and an anion exchange membrane (AEM; Fumasep[®] FAB): (a/b) hydrogen fraction in the gas collection chamber and cumulative acetate consumption; (c/d) calculated actual hydrogen production, expected hydrogen production based on the cumulative charge production, and the total gas production (Milligascounter[®]).

predicting the performance of a biocatalyzed electrolysis configuration.

The measured values for the total gas production (Milligascounter[®]) and hydrogen fraction in the gas collection chambers were used to calculate the actual hydrogen production according to Eq. (1). The expected hydrogen production based on the cumulative charge production was calculated according to Eq. (2). The cumulative values after 48 h for acetate consumption, charge production, and hydrogen production were used to calculate three process efficiencies (Table 1): (i) coulombic efficiency; (ii) cathodic hydrogen efficiency; and (iii) overall hydrogen efficiency. The overall hydrogen efficiencies were comparable for both single chamber configurations in this study and were around 23%. These overall hydrogen efficiencies were the result of low coulombic efficiencies and high cathodic hydrogen efficiencies.

3.2.1. Coulombic efficiency

In contrast to previous biocatalyzed electrolysis experiments, the coulombic efficiencies of both configurations were low. Liu et al. (2005b) achieved coulombic efficiencies of 60% to 78% and in our previous study in a similar electrochemical cell as used in this study we achieved a coulombic efficiency of $92 \pm 6.3\%$ (Rozendal et al., 2006b). The significant difference with our previous study was most likely caused by the fact that the electrochemical cells in our previous study were operated in batch mode on autoclaved medium, while both electrochemi-

Table 1 – Efficiencies^a of single chamber biocatalyzed electrolysis of acetate with a cation exchange membrane (CEM) and an anion exchange membrane (AEM)

	CEM (%)	AEM (%)
Coulombic efficiency (acetate to e ⁻)	22.8 ± 0.2	23.2 ± 1.5
Cathodic hydrogen efficiency (e ⁻ to H ₂)	101.4 ± 0.7	101.3 ± 0.6
Overall hydrogen efficiency (acetate to H ₂)	23.1 ± 0.3	23.5 ± 1.7

^a Calculations based on: 1 acetate → 8 e⁻ → 4 H₂.

cal cells in this study had been operated in continuous mode on non-autoclaved medium for over two months prior to the hydrogen yield tests. This difference in operation of the cells might have resulted in the growth of methanogenic biomass inside the electrochemical cell (Kim et al., 2005), which was also suggested by the occurrence of dark spots of biomass on the walls of the reactor. This methanogenic biomass competes with the electrochemically active micro-organisms for acetate, which decreases the coulombic efficiency. Limiting the growth of this biomass, therefore, needs to be a topic of further study to be able to increase the coulombic efficiency. Strategies to achieve this can be to increase the anode surface area to system volume ratio and to optimize the COD loading rate (Rabaey et al., 2005).

3.2.2. Cathodic hydrogen efficiency

The conversion of electrons to hydrogen (i.e. cathodic hydrogen efficiency) achieved in this study was practically complete in both configurations. This is in the same range as the 92% previously reported by Liu et al. (2005b) and much higher than the $57 \pm 0.1\%$ we achieved in our previous study in a similar electrochemical cell as used in this study (Rozendal et al., 2006b). In our previous study calculations have shown that a large part of the produced hydrogen at the cathode was lost by diffusion of hydrogen from the cathode into the anode chamber through the ion exchange membrane. In the same study, however, we predicted that the importance of this diffusional hydrogen loss reduces at increasing current densities. At higher current densities, i.e. at higher hydrogen production rates, this diffusional hydrogen loss remains more or less constant in absolute sense, but becomes less important in relative sense. This prediction has now been confirmed by the results of the hydrogen yield tests presented in this study. In our previous study the current density was about 0.5 A/m^2 (at 0.5 V), while in this study the current density was between 2.0 and 2.5 A/m^2 (at 1.0 V). Indeed, at these higher current densities the cathodic hydrogen efficiencies have increased to about 100%. This is an important fact for future studies, which should focus on achieving these higher current densities at lower applied voltages.

3.2.3. Volumetric hydrogen production rates

The average volumetric hydrogen production rates were comparable for both configurations. The CEM configuration produced $0.33 \text{ m}^3 \text{ H}_2/\text{m}^3$ reactor liquid volume/day and the AEM configuration produced $0.31 \pm 0.01 \text{ m}^3 \text{ H}_2/\text{m}^3$ reactor liquid volume/day. This is much higher than the $0.02 \text{ m}^3 \text{ H}_2/\text{m}^3$ reactor liquid volume/day that was achieved in our previous study in the two-chamber configuration (Rozendal et al., 2006b). However, in our previous study the hydrogen was produced at an applied voltage of 0.5 V , while in this study the hydrogen was produced at 1.0 V , because the performance of both configurations at an applied voltage 0.5 V was too low to do reliable measurements. This low performance at an applied voltage 0.5 V was caused by various potential losses in the system (see below).

3.3. Electrochemical impedance spectroscopy

The electrochemical cells performing biocatalyzed electrolysis of acetate were subjected to EIS at 1.0 V (Fig. 4). Interpretation of the Nyquist plots that were obtained through EIS was limited to the determination of the ohmic resistance of the system according to the methodology as applied by He et al. (2006). As can be seen from the Nyquist plots both configurations show a comparable ohmic resistance in the order of 2Ω . Although these values seem low compared to the values reported in other studies (He et al., 2006; Min et al., 2005), it is actually already too high for practical application. The average currents during the 48 h hydrogen yield tests at 1.0 V were 95 mA (2.39 A/m^2) for the CEM configuration and 86 mA (2.15 A/m^2) for the AEM configuration, which means that the ohmic losses at these values already amount to 0.21 V ($= 95 \times 10^{-3} \text{ A} \times 2.2 \Omega$) for the CEM configuration and 0.15 V ($= 86 \times 10^{-3} \text{ A} \times 1.8 \Omega$) for the AEM configuration. This is a significant part of the total applied voltage of 1.0 V . In our system a substantial part of the ohmic resistance is caused by the anode itself. EIS of only the anode at a potential of -0.25 V showed that the anode ohmic resistance was 0.64Ω in the CEM configuration and 0.75Ω in AEM configuration. This resistance is believed to be caused largely by the contact resistance of the graphite fibers in the graphite felt.

3.4. Permeated water analysis

During the hydrogen yield tests small amounts of water permeated through the membrane from the anode to the cathode side of the membrane into the gas collection chamber. Ions that are transported through the membrane will accumulate in this permeated water. Table 2 shows that less water permeated through the AEM than through the CEM. In both configurations the permeated water had an increased pH as compared to the anode chamber (pH 6.8–7; conductivity: 5.22 mS/cm). As was predicted beforehand, the AEM is capable of significantly reducing the pH increase at the cathode, but the effect was smaller than anticipated. In both configurations the pH increase is caused by the membrane

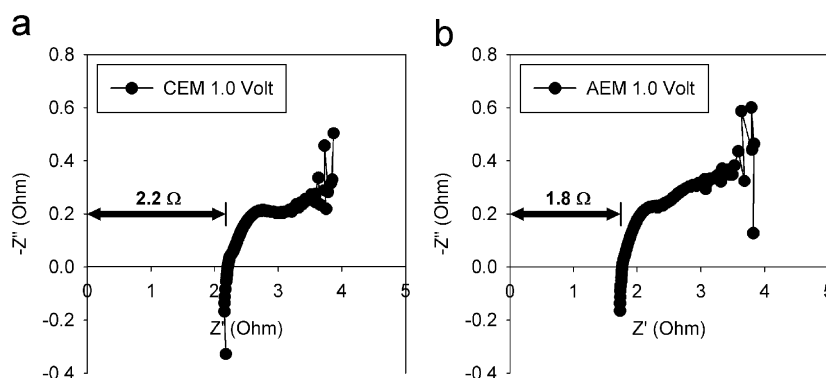


Fig. 4 – Electrochemical impedance spectroscopy (EIS): Nyquist plots of single chamber biocatalyzed electrolysis of acetate at an applied voltage of 1.0 V with: (a) a cation exchange membrane (CEM; Nafion® 117); and (b) an anion exchange membrane (AEM; Fumasep® FAB). The arrows indicate the ohmic resistance in the system.

Table 2 – Permeated water volume, composition and calculated H^+/OH^- transport number ($t_{H/OH}$) after the 48 h hydrogen yield tests

	CEM	AEM
Volume (ml)	33.5	15.5
pH (dimensionless)	13.2	11.1
Conductivity (ms/cm)	73.05	3.06
Na ⁺ (mM)	125.7	7.1
K ⁺ (mM)	170.3	12.3
NH ₄ ⁺ (mM)	8.8	3.5
Cl ⁻ (mM)	0.6	0.2
HCO ₃ ⁻ /CO ₃ ²⁻ (mM)	4.8 ^a	11.9 ^b
Acetate (mM)	0.38	0.02
$t_{H/OH}$ (dimensionless)	0.94	1.00

^a At pH 13.2 $HCO_3^-:CO_3^{2-} = 0.001:1$.
^b At pH 11.1 $HCO_3^-:CO_3^{2-} = 0.17:1$.

transport of ions other than protons and hydroxyl ions. Permeated water analysis showed that in both configurations others ions than protons and hydroxyl ions were transported through the membrane during the hydrogen yield test. However, this transport was much smaller in the AEM configuration as can be seen from the ion concentrations, the conductivity of the permeated water, and the calculated transport number $t_{H/OH}$. The transport of ions other than protons and hydroxyl ions is caused by the fact that ion exchange membranes are never 100% selective for protons or hydroxyl ions and, moreover, not even for only cations or anions.

The concentrations of ions other than protons and hydroxyl ions can be used to predict pH of the permeated water by assuming that for every net positive charge in the form of others ions than protons and hydroxyl ions transported into the permeated water, one hydroxyl ion accumulates in the permeated water. For the permeated water in the CEM configuration this predicts a pH of 13.5; for the permeated water in the AEM configuration a pH of 10.7. This prediction confirms the dramatic pH effects of the transport of only small amounts of ions other than protons and hydroxyl ions. This dramatic pH effect especially becomes clear when studying the composition of the permeated water of the AEM configuration. During the 48 h hydrogen yield tests, low amounts of ions other than protons and hydroxyl ions were transported to the permeated water and the transport number in the AEM configuration was almost equal to unity. Still, the pH effect was substantial.

Unfortunately, an extra 0.06 V potential loss occurs for every pH unit increase of the permeated water at the cathode as compared to the anode pH (CEM $\Delta pH = 6.4$; AEM $\Delta pH = 4.4$). A large part of the cathode/membrane overpotentials observed in the applied voltage scans, therefore, were likely caused by these pH gradient related potential losses. Furthermore, these results also indicate how difficult it is to prevent the pH gradient related potential losses, especially in the case of single chamber configurations. The ion concentrations increase more rapidly in a small amount

of permeated water in single chamber configurations than in the total liquid content of a cathode chamber in two-chamber configurations. Therefore, the pH increase at the cathode and the pH gradient related potential losses will occur more rapidly in single chamber configurations as compared to a two-chamber configurations (Rozendal et al., 2006a). This also explains why the performance at an applied voltage of 0.5 V was so low in this study, as the applied voltage was too low to overcome the high pH gradient related potential losses of the single chamber configuration.

The obtained results suggest that the pH gradient related potential losses seem to be a fundamental problem associated with the use of ion exchange membranes in single chamber bio-electrochemical systems running on wastewater. In MFCs, omitting the membrane seems to be a solution, as was demonstrated by Liu and Logan in a single chamber MFC study (Liu and Logan, 2004). For biocatalyzed electrolysis, however, omitting the membrane would result in a contamination of the hydrogen product gas and in a reduction of the coulombic efficiency due to consumption of the hydrogen in the anode chamber.

3.5. Scanning electron microscopy

The MEAs that were used in the experiments were studied using scanning electron microscopy (SEM) combined with element analysis (Fig. 5). The element analysis showed that the electroless plating procedure was more effective for the CEM than for the AEM, as the platinum profile is higher and can be found deeper into the CEM than into AEM. This has made the catalytic region larger in the CEM MEA than in the AEM MEA. Furthermore, the platinum peak close to the surface is much higher in the CEM platinum profile than in the AEM platinum profile. According to Millet et al. (1990) this peak represents a conducting platinum layer outside the membrane, which is responsible for the electrical contact of the catalytic platinum microparticles inside the membrane with the electrical circuit. If this conducting platinum layer outside the membrane is not well developed during the electroless plating procedure, the catalytic platinum microparticles become isolated and inactive for hydrogen production. The difference in visual appearance of the MEAs also was an indication for this. Whereas the outside of the CEM MEA developed a metallic appearance during the electroless plating procedure, the outside of the AEM MEA only turned black. The AEM MEA, therefore, might have lacked a well developed conducting platinum layer outside the membrane with inactive catalytic platinum microparticles inside the membrane as a result. A lower activity of catalytic platinum microparticles leads to increased cathode overpotentials (see below).

3.6. Potential losses

Table 3 gives an overview of the distribution of the applied voltage of 1.0 V during the 48 h hydrogen yield tests for both single chamber biocatalyzed electrolysis configurations. As was predicted by the analysis of the permeated water, the pH gradient associated potential loss, caused by the membrane charge transport of ions other than protons and hydroxyl

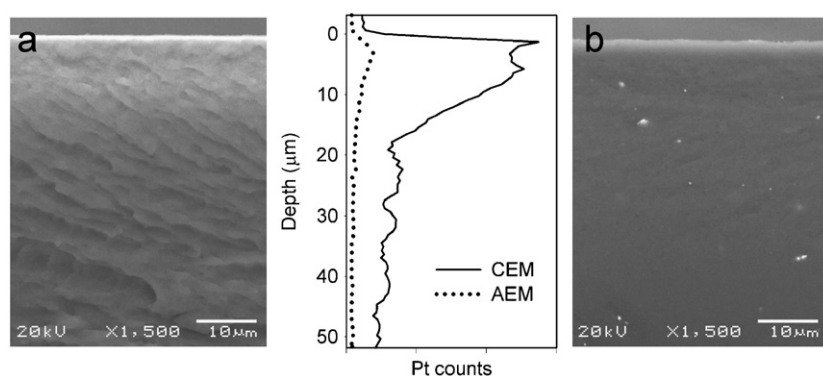


Fig. 5 – SEM photos of a cross section of the membrane electrode assemblies (MEAs) used in the single chamber biocatalyzed electrolysis experiments: (a) MEA with a cation exchange membrane (CEM; Nafion[®] 117); (b) MEA with an anion exchange membrane (AEM; Fumasep[®] FAB). The graph in the middle shows the platinum profiles inside the membranes resulting from the electroless plating procedure.

Table 3 – Overview of the potential losses during the 48 h hydrogen yield tests at an applied voltage of 1.0 V

	CEM (at 2.39 A/m ²)	AEM (at 2.15 A/m ²)
Equilibrium voltage (V)	0.17 ^a	0.17 ^a
Ohmic loss (V)	0.21 ^b	0.15 ^b
Membrane pH gradient loss (V)	0.38 ^c	0.26 ^c
Anode overpotential (V)	0.12 ^d	0.15 ^d
Cathode overpotential (V)	0.12 ^e	0.27 ^e

^a Calculated for 1 mM acetate; 1 bar H₂.
^b Calculated from the ohmic resistances measured in the Nyquist plots (Fig. 4).
^c pH gradient across the membrane (CEM ΔpH = 6.4; AEM ΔpH = 4.4) multiplied by 0.06 V.
^d Values corrected for the ohmic loss of the anode itself, as this was already included in the total ohmic loss (the ohmic resistance of the anode was determined with EIS: CEM 0.64 Ω; AEM 0.75 Ω).
^e Cathode overpotential = applied voltage – equilibrium voltage – ohmic loss – membrane pH gradient loss – anode overpotential.

ions, forms a significant part of the total potential losses in both configurations. However, Table 3 also shows that other potential losses are not insignificant either.

Even though the ohmic resistance is low compared to other studies (He et al., 2006; Min et al., 2005), the ohmic potential loss is large. This paradox can be explained from the larger electrochemical cells that have been used in this study as compared to other studies. Consequently, the electrical currents that flow through the cells are larger and therefore size independent resistances (e.g. contact resistances) cause larger ohmic losses than the same resistances in smaller systems. This shows the importance of also studying larger systems in order to evaluate the consequences of scale-up (Logan et al., 2006).

The anode overpotentials are in the same range as those observed in other studies with acetate as the electron donor (Liu et al., 2005a; Ter Heijne et al., 2006). They were similar in both configurations, which is not surprising as the conditions

in the anode chambers were comparable. The anode overpotentials represent the smallest losses in the system, but at the relatively large current densities of around 2 A/m² they also have started to become significant. A possible reason for the increased anode overpotentials is the relatively low anode acetate concentration of around 1 mM during the hydrogen yield tests. Further, as the system was not stirred vigorously, concentration polarization could have started to occur (Logan et al., 2006). One of the effects that can take place as a result of concentration polarization is the occurrence of low pH near the anode due to proton production. In future studies, the application of granular graphite (Rabaey et al., 2005; Sell et al., 1989) might help to lower the anode overpotentials.

The cathode overpotentials represent an important part of the total potential losses. In the AEM configuration the cathode overpotentials were higher than in the CEM configuration. One issue that needs to be examined further in this respect is the effectiveness of the MEA preparation method used for the AEM. SEM combined with element analysis showed that the penetration depth of the platinum was smaller in case of the AEM configuration and that the conducting platinum layer outside the membrane was less developed. This probably caused the difference in catalytic activity of MEAs and resulted in increased cathode overpotentials for the AEM MEA. The MEA preparation method was an adapted electroless plating method, originally developed for CEMs. In order to optimize the AEM MEA performance the influence of the plating parameters needs to be examined (e.g. increasing exposure time to the Pt-solution or the sodium borohydride). Also other MEA preparation methods can be examined in future studies.

3.7. Outlook

In this study the electrical energy requirement for hydrogen production through biocatalyzed electrolysis was about 2.2 kWh/m³ H₂ in both configurations. This is much lower than the 4.4–5.4 kWh/m³ H₂ required by commercial water electrolyzers (Turner, 2004). Our goal for future studies is to further reduce the electrical energy requirement of biocatalyzed electrolysis to below 1.0 kWh/m³ H₂. This implies that the applied voltage needs to be lowered to at least

0.46 V. At the same time we aim at increasing the volumetric hydrogen production rate from 0.3 (this study) to about 10 m³ H₂/m³ reactor liquid volume/day. Increasing the volumetric hydrogen production rate means increasing the current density of the system. Based on recent results achieved in our laboratories (Ter Heijne et al., 2006), we estimate that an optimized biological anode is capable of achieving current densities of between 5 and 10 A/m². At these current densities we can afford only 0.29 V of potential losses (= applied voltage–equilibrium voltage = 0.46–0.17 V). To achieve these goals major improvements of the electrochemical and geometrical design of the biocatalyzed electrolysis system are required.

First, this can only be achieved if the pH gradient associated potential losses as observed in this study can be eliminated completely in an optimized design of the biocatalyzed electrolysis system. For this purpose, the interesting characteristics of the AEM in this respect need to be further studied or other means of reducing the pH gradient associated potential loss need to be explored (e.g. other types of membranes).

Ohmic losses must be reduced by minimizing the electrode spacing and choosing better electrode materials (Logan et al., 2006). In order to reduce the ohmic loss of the system to 0.05–0.10 V at 5–10 A/m², the total ohmic resistance of the system needs to be reduced to between 50 and 200 Ωcm². For comparison, total ohmic resistances of water electrolyzers are typically in the order of 1 Ωcm² (Kinoshita, 1992).

The overpotential of HER cathodes in water electrolyzers can be as low as 0.025 V at 10000 A/m² (Kinoshita, 1992), which indicates that there is much room for optimization of HER cathodes in biocatalyzed electrolysis. Assuming that both the ohmic loss and the cathode overpotential can indeed be reduced to 0.05–0.10 V, anode overpotentials need to be reduced to 0.09–0.19 V. Extrapolating recent results achieved in our laboratories (Ter Heijne et al., 2006), we believe that achieving these kinds of anode overpotentials at current densities of 5–10 A/m² is realistic, but will require much fundamental research into the working principles of biological anodes.

Finally, an optimized system will also require an optimized geometrical design. The biocatalyzed electrolysis system that was used in this study, operating at a volumetric hydrogen production rate of 0.3 m³ H₂/m³ reactor liquid volume/day (at about 2 A/m²), has an anode surface area to system volume ratio of 12 m²/m³ reactor liquid volume. In order to reach the aimed volumetric production rate of 10 m³ H₂/m³ reactor liquid volume/day (at 5–10 A/m²), the anode surface area to system volume ratio will have to increase to about 80–160 m²/m³ reactor liquid volume. Similar anode surface area to system volume ratios have been demonstrated previously in MFC studies (Kim et al., 2002). Applying them also to biocatalyzed electrolysis brings the process one step closer to practical application.

4. Conclusions

1. It is possible to operate biocatalyzed electrolysis in single chamber configuration, although in this study the performance at voltages below 0.6 V was limited.
2. It is possible to operate single chamber biocatalyzed electrolysis both with a CEM in the configuration as with an AEM in the configuration. Both configurations demonstrate a comparable performance, producing over 0.3 m³ H₂/m³ reactor liquid volume/day at 1.0 V applied voltage.
3. The coulombic efficiencies of both biocatalyzed electrolysis configurations were low (around 23%), which was most likely caused by the activity of methanogenic bacteria in the anode chamber. The cathodic hydrogen efficiencies, on the other hand, were high (close to 100%) in both configurations, which was the result of the high current densities (between 2.0 and 2.5 A/m²). The resulting overall hydrogen efficiencies for both single chamber configurations were around 23%.
4. Analysis of the water that permeated through the membrane revealed that a large part of potential losses in the system are associated with a pH gradient across the membrane. This pH gradient is caused by the membrane charge transport in the form of ions other than protons and hydroxyl ions. These pH gradient related potential losses are more problematic for single chamber configurations than for two-chamber configurations, as ion concentrations increase more rapidly in a small amount of permeated water in a single chamber configuration than in the total liquid content of a cathode chamber in a two-chamber configuration.
5. The AEM is better capable of preventing the pH gradient across the membrane than the CEM (CEM ΔpH = 6.4; AEM ΔpH = 4.4) as a result of its alternative ion transport properties. Consequently, the pH gradient associated potential losses were lower in the AEM configuration (CEM 0.38 V; AEM 0.26 V).
6. In the AEM configuration, on the other hand, the reduced pH gradient associated potential losses were counteracted by the increased cathode overpotentials as compared to the CEM configuration (CEM 0.12 V at 2.39 A/m²; AEM 0.27 V at 2.15 A/m²). SEM combined with element analysis suggested that the increased cathode overpotentials in the AEM configuration were the result of a less effective electroless plating method for the AEM MEA.

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